

Auray, Devereux, and Eyquem: “Tariffs, Automation, and Business Dynamism”

Discussion by Lukas Freund

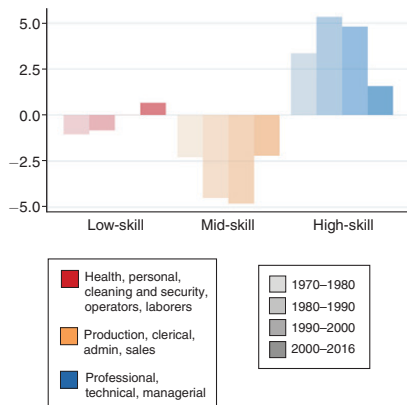
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Motivation: labor market polarization trend & shift toward tariffs

Automation → labor market polarization



U.S. tariff policy

“Jobs and factories will come roaring back into our country”

*D. Trump’s “Liberation Day” remarks
(April 2, 2025,)*

FIGURE 4. CHANGES IN OCCUPATIONAL EMPLOYMENT SHARES AMONG WORKING-AGE ADULTS, 1970–2016

Source. Autor (2019).

This paper

- **Question:** Can tariff barriers slow automation, preserve routine jobs?
 - can they revive business dynamism?
- **Model:** trade *[Ghironi-Melitz, 2005]* w/ task-based production *[Acemoglu/Autor/Restrepo/...]*
 - standard trade channel: $T \uparrow \Rightarrow p_{\text{imports}} \uparrow$, RER appreciates, $IM \downarrow$ $EX \downarrow$
 - entry channel: domestic market value $\uparrow \Rightarrow$ entry $\uparrow \Rightarrow n_{\text{firms}} \uparrow$, output per firm $\downarrow$
 - **automation channel:** $mc \downarrow$ while robot price ϕ is fixed $\Rightarrow \phi/mc \uparrow \Rightarrow a \downarrow$
- **Evidence** from the 2018–2019 U.S. tariff episode
- $\rightarrow$ **Answer:** **unilateral tariffs...**
 - ...can temporarily boost entry and number of domestic varieties
 - **...can slow down automation, persistently shifting production toward routine labor**
 - ...cannot generate a sustained expansion of factor-price output

Discussion

Contributions...

- **...to literature:** integrate trade & automation models → can study how production structure is affected by tariffs

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- **...to policy discussion:** provocative!
 - policy claim: tariffs can bring production and manufacturing jobs home
 - standard objection: won't work, decline in routine jobs mostly reflects automation
 - this paper: tariffs protect old industries from technology-induced task displacement

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Contributions...

- **...to literature:** integrate trade & automation models → can study how production structure is affected by tariffs
- **...to policy discussion:** provocative!
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 - this paper: tariffs protect old industries from technology-induced task displacement

Comments.

- ① Theory: is holding the price of robots fixed reasonable? what if we don't?
- ② Empirics: does the evidence support the model mechanism?
- ③ Interpretation: are effect sizes meaningful? “business dynamism”?

Comment 1: Review of the theoretical mechanism

- **Equilibrium:** given mc, ϕ
 - robots & workers are perf. substitutes in R tasks + interior $a_t \rightarrow w_t^u = \psi_t \phi_t$ [eq.'s 17 + 18]
 - automated task share a_t is a decreasing function of ϕ_t/mc_t [eq. 21]
 - non-routine wage: $w_t^h \propto mc_t$ [eq. 16]
- **Transmission:** tariff $\Rightarrow$ RER appreciates ($\hat{q} < 0$) $\Rightarrow$ producer prices fall $\Rightarrow mc_t \downarrow$
 - *quantities* adjust via robots
 - *prices* adjust via w^h (protection cannot raise w^u in *any* experiment given fixed ϕ)

Automation slows *if* ϕ_t falls by less than mc_t & paper *assumes* ϕ_t is fixed.

Comment 1: Treating robot cost as fixed is key

- **Key modeling choice:** paper takes robot cost ϕ to be fixed in Home CPI units ($\hat{\phi} = 0$)
- **Why?** Cost of robots isn't derived from a supply block nor world-determined
 - resource cost of robots? robot production absorbs no labor, no intermediates
 - “world-determined”? experiments fix ϕ_t in Home CPI units *and* ϕ_t^* in Foreign CPI units, while $\hat{q} \neq 0$; LOP would require $\hat{\phi} = \hat{q} + \hat{\phi}^* \rightarrow 2$ local anchors $\neq$ one world price $\rightarrow$ robots *de facto* nontraded
- **Alternative treatments of robot price ϕ are interesting & plausible**
 - $\rightarrow$ re-solve static example with $\hat{\phi}$ left free, close it under 3 alternative ‘models’ for ϕ

Comment 1: 3 alternative models, 3 different predictions

- Case (a) - derive $\hat{\phi} = 0$ by modeling robots as produced from H cons. bundle
 - $\leftrightarrow$ seems implausible for robot production? & inconsistent with rest of paper
- Case (b) – produced with intermediate good $\rightarrow$ no automation effect
- Case (c) – **imported**: appreciation cheapens robots $\rightarrow$ signs flip
 - consistent with US industrial robots mostly imported from Japan + Europe [IFR reports]

What is a robot made of?	$\hat{\phi}$	Automation	Skill premium	R wage	R emp.
(a) Home consumption basket	0	$\downarrow$ $-\frac{\chi a^T}{4\alpha\omega_\ell\Delta}$	$\downarrow$ $-\frac{T}{4\alpha\Delta}$	$-$ 0	$\uparrow$ $+\frac{\kappa T}{4\alpha\omega_\ell\Delta}$
(b) Home intermediate good	$\widehat{mc}$	0	0	$\downarrow$ $-\frac{T}{4}$	0
(c) Made abroad/world price	$\hat{q} = -\frac{T}{2} < 0$	$\uparrow$ $+\frac{\chi a^T}{4\alpha\omega_\ell}$	$\uparrow$ $+\frac{T}{4\alpha}$	$\downarrow$ $-\frac{T}{2}$	$\downarrow$ $-\frac{\kappa T}{4\alpha\omega_\ell}$

Comment 1: Scenarios that recover the de-automation result

- **When *do* tariffs lower automation?**

- ① Over short horizons, capital-goods imports USD-invoiced, FX pass-through low $\rightarrow \hat{\phi} \approx 0$
 $\leftrightarrow$ but paper focuses on a permanent fall
- ② **Robots are imported and themselves tariffed** $\rightarrow$ de-automation once the effective robot tariff is sufficiently high
 $\leftrightarrow$ in 2018, Europe+Japan (main suppliers) untaxed – but 2025?
- ③ **Symmetric retaliation:** the FX rate doesn't move, while producer prices still fall (both countries' tariffs tax trade) $\rightarrow$ automation $\downarrow$
 - maybe retaliation extension deserves more prominence?

Whether tariffs slow automation depends on how the robot price is determined.

Comment 2: Relating the empirical exercise to the model

► Entry & BDS timing

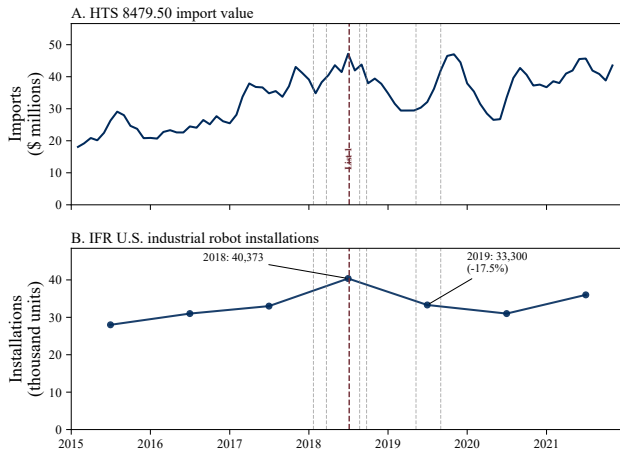
- **Exercise:** compare CZs with different pre-tariff industry composition (share) given 2018/2019 tariff exposure (shift)

$$\text{EXP}_c = \sum_i \frac{L_{ci,2017}}{\sum_j L_{cj,2017}} \text{EXP}_i, \quad \text{EXP}_i \in \{\text{Cost}_i, \text{Protect}_i, \text{Retal}_i\}$$

- **Finding:** CZ cost-side exposure predicts shift toward routine production tasks
→ “supports the model’s task-reallocation [...] mechanism”
- **Comments:**
 - 1 Model’s mechanism is an agg. relative price → differenced out by construction in cross-sectional shift-share design
 - 2 Changes reflect *industry mix*, with *fixed* task scores → no establishment alters technology adoption/task allocation

It would be valuable to align the empirical exercise more closely with the model.

Comment 2: Suggestive aggregate evidence from IFR & import data



Notes. Panel A: monthly U.S. import value of HTS 8479500000 (“industrial robots”), Census customs data. Panel B: annual U.S. International Federation of Robotics (IFR) industrial-robot installations.

Coda: Paper's research question is highly relevant in 2026

- Automation discussion in 2026 is all about **AI – trade & tariffs play a big role!**
→ reinforces importance of modeling tariff exposure of automation inputs

(a) Surge in imports of AI-related products

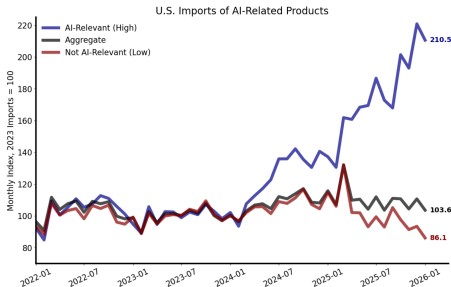
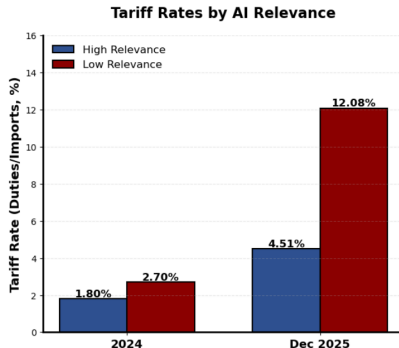


Figure 1: U.S. Imports of AI and non-AI products

(b) ...which face lower effective tariffs



Source. Waugh (2026). AI-related products: products used in construction and operation of AI infrastructure.

Summary

Provocative & timely paper that challenges conventional discourse on the principal mechanism through which tariffs affect production and labor demand.

- **Suggestions:**

- ① Explicitly model cost of machines as traded goods
- ② Align empirical analysis more closely with model mechanism
- ③ Focus on automation margin rather than “business dynamism”

Thank You!

Extra Slides

Extra comment: What do the model results tell us about the research question?

- Taking model results at face value, what do they imply for the research question(s)?
- **“Can protectionism slow automation?”**
 - a permanent 10pp tariff lowers automated task share by ~ 1 pp at 40yrs
 - task displacement $\uparrow 14\text{--}15$ pp over 1984–2016 ≈ 0.45 pp per year [Acemoglu-Restrepo, 2022]
 - headline experiment sets the trend back by ~ 2 yrs

⇒ **The tariff is a meaningful but small lever on automation given a strong trend**
- **“Can protectionism revive business dynamism?”**
 - shutting down varieties barely changes the output response
 - model features no job reallocation, no firm growth, innovation, etc. = the classic margins of the macro literature cited [Decker et al., 2016; Akcigit-Ates, 2023; ...]

⇒ **Unclear – but studying the automation margin is (imho) ambitious enough**

Extra comment: Dating the BDS entry results



Table 3: "2017–2018"
Difference ending in Mar. 2018

$$\hat{\beta}_{\text{cost}} = +2.82^{**}$$

Protection-side: -0.76

Table 3: "2018–2019"
Difference ending in Mar. 2019

$$\hat{\beta}_{\text{cost}} = -3.71^{***}$$

Protection-side: $+1.09^*$

- Paper: more tariff-exposed places saw business creation $\uparrow$ 2017–18 and $\downarrow$ 2018–19.
- What does BDS data mean by "2018"? BDS: growth rates are based on March-to-March changes; tabulations for a given year are the changes from the prior year to the current year.
- **Implication:** measured entry increase predates the tariffs' implementation; entry fell where cost-side exposure hit during the tariff-active year *[consistent with*